

Introduction to Rings.

A ring R is a set with two binary operations $+$ (addition), and $\cdot$ (usually multiplication) satisfying the following:

- 1) $(R, +)$ is an abelian group
- 2) $\cdot$ is associative (i.e. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$)
- 3) Distributive laws:
$$\left. \begin{aligned} (a+b) \cdot c &= ac + bc \\ a(b+c) &= ab + ac \end{aligned} \right\} \forall a, b, c \in R.$$

Commutative: $a \cdot b = b \cdot a \quad \forall a, b \in R.$

Identity: $1 \in R$ with $1 \cdot a = a \cdot 1 = a \quad \forall a \in R.$

Remark: Suppose $1 \in R$. Then the commutative under addition is forced by the distributive laws.

(i.e. Dis. laws $\Rightarrow$ Comm. under addition).

$$(1+1)(a+b)$$

$$= 1(a+b) + 1(a+b)$$

$$= 1 \cdot a + 1 \cdot b + 1 \cdot a + 1 \cdot b$$

$$= a + b + a + b.$$

$$(1+1)(a+b)$$

$$= (1+1)a + (1+1)b$$

$$= 1 \cdot a + 1 \cdot a + 1 \cdot b + 1 \cdot b$$

$$= a + a + b + b.$$

By left/right cancellation $\boxed{a+b = b+a.}$

Division ring: A ring with 1 (where $1 \neq 0$) is called a division ring (or skew field) if every non-zero element $a \in R$ has a multiplicative inverse.

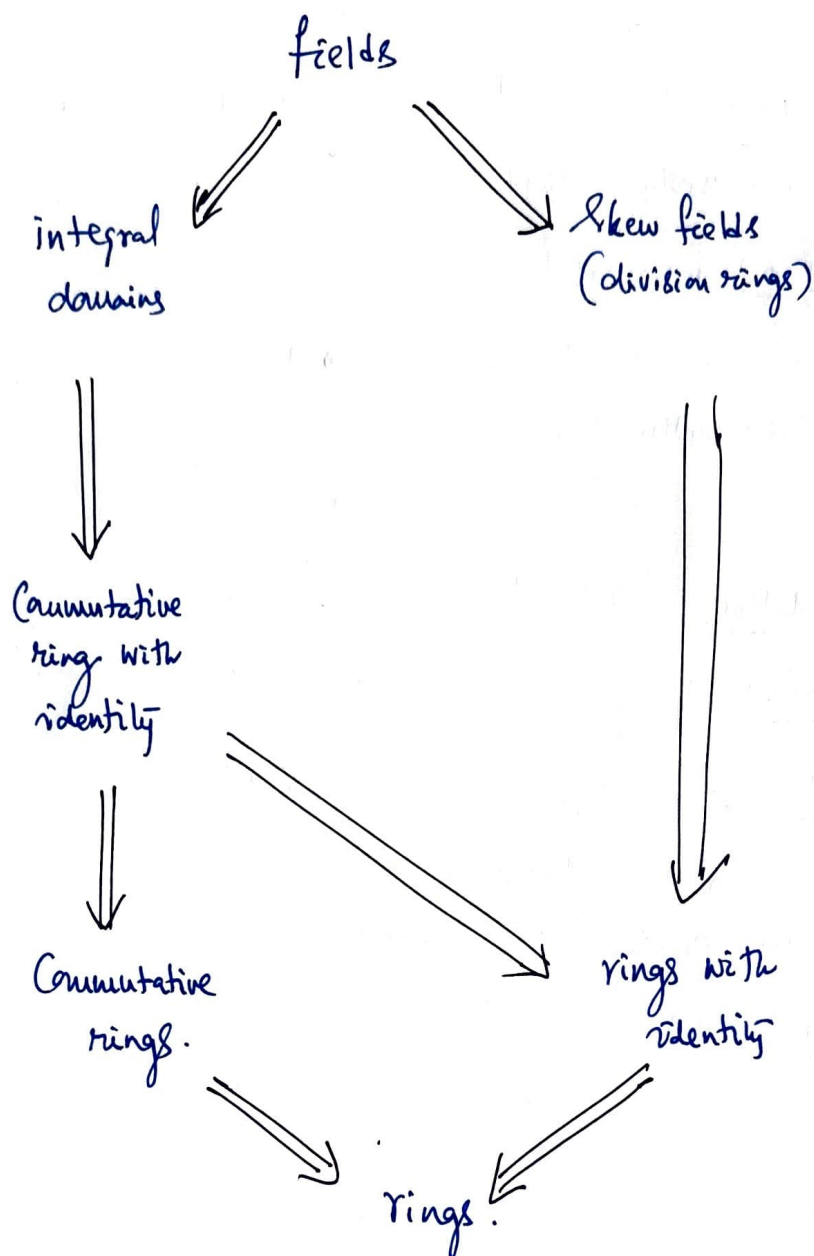
Field: A commutative division ring is a field.

Ex. 1) Trivial ring: $(R, +)$ is an abelian gp.

Define $\cdot$ by $a \cdot b = 0 \quad \forall a, b \in R$

2) $(\mathbb{Z}, +, \cdot)$ Commu. ring with 1

3) $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$ Commu. rings with 1 (also fields)

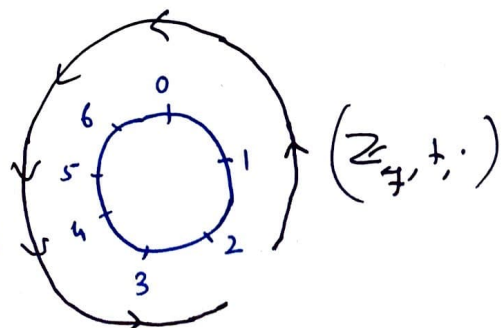


Properties: Let R be a ring. Then.

- 1) $0a = a0 = 0 \quad \forall a \in R$
- 2) $(-a)b = a(-b) = -(ab) \quad \forall a, b \in R$
- 3) $(-a)(-b) = ab$
- 4) If $1 \in R$, then $-a = (-1)a \quad \forall a \in R$.

Ex: (ring/field) $\cdot (\mathbb{Z}_7, +, \cdot)$

$$\begin{aligned}
 2 - 6 &= 2 + (-6) \\
 &= 2 + 1 \quad (-6 \equiv 1 \pmod{7}) \\
 &= 3
 \end{aligned}$$



Ex: Consider the set (Field with 9 elements).

$$\mathbb{Z}_3[i] = \{a + bi \mid a, b \in \mathbb{Z}_3\}, \quad i = \sqrt{-1}, i^2 = -1$$

$$= \{0, 1, 2, i, 1+i, 2+i, 2i, 1+2i, 2+2i\}$$

9 elements.

$$(1+2i) * (2+i) = 1 \cdot 2 + 1 \cdot i + 2i \cdot 2 + 2i \cdot i$$

$$= 2 + 5i + 2 \cdot i^2 = 2 + 5i + 2(-1)$$

$$= 0 + 5i$$

$$= 2i \quad (\text{since } 5 \equiv 2 \text{ in } \mathbb{Z}_3).$$

$$(1+2i) + (2+i) = 3 + 3i = 0 + 0i$$

$$= 0 \quad (\text{since } 3 \equiv 0 \text{ in } \mathbb{Z}_3).$$

Note: $-1 \equiv 2 \text{ in } \mathbb{Z}_3$.

Multiplication Table $\mathbb{Z}_3[i]^* = \mathbb{Z}_3[i] - \{0\}$.

	1	2	i	1+i	2+i	2i	1+2i	2+2i
1	1	2	i	1+i	2+i	2i	1+2i	2+2i
2	2	1	2i	2+2i	1+2i	1	2+i	1+i
i	i	2i	$-1 \equiv 2$				1+i	1+2i
1+i	1+i	2+2i	2+i	2i	1	1+2i	2	i
2+i	2+i		2+2i		i		2i	2
2i	2i	i	1	1+2i	1+i	2	2+2i	2+i
1+2i	1+2i	-	-	-	2i			2+i
2+2i	2+2i	1+i	1+2i	i	2	2+i	1	2i

Properties: Let R be a ring. Then

1) $0a = a0 = 0 \quad \forall a \in R$

2) $(-a)b = a(-b) = -(ab) \quad \forall a, b \in R$

3) $(-a)(-b) = ab$

4) If $1 \in R$, then the identity is unique, and $-a = (-1)a$

$$\begin{aligned}
 1) \quad 0a &= (0+0)a \\
 \Rightarrow 0+0a &= 0a+0a \\
 \Rightarrow 0 &= 0a \quad (\text{right cancellation})
 \end{aligned}$$

$$\begin{aligned}
 2) \quad ab + a(-b) &= a(b+(-b)) \\
 &= a(0) \\
 &= 0 \quad (\text{by (1)}) \\
 \Rightarrow a(-b) &= -(ab)
 \end{aligned}$$

$$\begin{aligned}
 3) \quad (-a)(-b) &= -((-a)b) \quad (\text{by (2)}) \\
 &= -(-(ab)) \quad (") \\
 &= ab.
 \end{aligned}$$

Def. (unit): Let R be a ring. An element $u \in R$ is called a unit if it has a multiplicative inverse $u^{-1} \in R$ such that $u^{-1}u = 1 (= uu^{-1})$.

Def. (Zero-divisor): Let R be a ring. A non zero element $a \in R$ is called a zero divisor if there exists a non zero $b \in R$ such that $a \cdot b = 0$.

Ex: In $(\mathbb{Z}_{10}, +, \cdot)$ $3 \cdot 7 = 1$ ($21 = 1$ in $\mathbb{Z}_{10}$) (units).
 $\Rightarrow 3^{-1} = 7$ and $7^{-1} = 3$.

Also, $5 \cdot 2 = 0$, $6 \cdot 5 = 0$ (Zero divisors)

Ex: Consider the ring: $(\mathbb{Z}_{12}, +, \cdot)$

Units: $\{1, 5, 7, 11\}$ (Since $\begin{aligned}
 1 \cdot 1 &= 1 \\
 5 \cdot 5 &= 1 \\
 7 \cdot 7 &= 1 \\
 11 \cdot 11 &= 1
 \end{aligned}$
} self inverse

Zero divisors: $\{2, 3, 4, 6, 8, 9, 10\}$
 (0 always a zero divisor) $\left. \begin{aligned}
 2 \cdot 6 \\
 3 \cdot 4 \\
 8 \cdot 9 \\
 6 \cdot 10
 \end{aligned} \right\} = 0 \quad (\text{product zero})$

Ex: Take $(\mathbb{Z}_7, +, \cdot)$

The units are: $\{1, 2, 3, 4, 5, 6\}$

Zero divisors: $\{0\}$ (no non zero-zero divisors).

Ex: Take $(\mathbb{Z}, +, \cdot)$. Units: 1 and -1.

Ex. In ring $(\mathbb{R}, +, \cdot)$.

Units: All non zero elements.

Zero divisors: There are no zero divisors (except '0').

Property: Let α be a non zero ring element. Then α cannot be both a unit and a zero divisor.

Verification: Let R be a ring, and suppose $\alpha \in R, \alpha \neq 0$, is both a unit and a zero divisor.

$$\Rightarrow \exists \alpha^{-1} \in R \exists \alpha^{-1} \cdot \alpha = 1$$

$$\exists \beta \in R \exists \alpha \cdot \beta = 0, \beta \neq 0$$

Now multiplying with (left side) α^{-1} both sides of $\alpha \cdot \beta = 0$.

$$\text{ie; } \alpha^{-1}(\alpha \cdot \beta) = \alpha^{-1} \cdot 0$$

$$\Rightarrow (\alpha^{-1} \cdot \alpha) \beta = \alpha^{-1} \cdot 0$$

$$\Rightarrow 1 \cdot \beta = 0; \text{ a contradiction.}$$

Therefore such α does not exist.

Def: (Subring): Let R be a ring. A subset S of R is a subring of R if S is a ring using the operations of R .

Ex. 1) $\mathbb{Z}$ is a subring of $\mathbb{R}$.

2) $\mathbb{Q}$ is a subring of $\mathbb{R}$.

3) $S = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} \mid a, b \in \mathbb{Z}_{10} \right\}$ is a subring of $M(\mathbb{Z}_{10})$.

$$a, b \in S \Rightarrow a+b \in S$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \text{additive identity.}$$

$$a \in S \Rightarrow -a \in S; \text{ ie, } \begin{pmatrix} a & b \\ b & a \end{pmatrix} \in S \Rightarrow \begin{pmatrix} -a & -b \\ -b & -a \end{pmatrix} \in S.$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} : \text{multiplicative identity.}$$

Def: (Group of units): Let R be a ring. Then the subset

$R^* = \{a \in R \mid a \text{ is a unit}\}$ is called the group of units of R .

Ex. For each ring R , find the group of units R^* .

1) $R = \mathbb{Z}_{10}$, $R^* = \{1, 3, 7, 9\} = U(10)$.

2) $R = M(\mathbb{Z}_{10})$, $R^* = G(\mathbb{Z}_{10}) = \text{units in } M(\mathbb{Z}_{10})$
 $= \{ \alpha \in M(\mathbb{Z}_{10}) \mid \alpha \text{ is a unit} \}.$

3) $R = \mathbb{Z}$, $R^* = \{1, -1\}$

4) $R = \mathbb{R}$, $R^* = \mathbb{R} \setminus \{0\}$.

Property: Let R be a ring and define $R^* = \{a \in R \mid a \text{ is a unit}\}$.
Then R^* is a multiplicative group.

Hint: product of two units is a unit.

Def. (Integral domain). A commutative ring with identity ($1 \neq 0$) is called an integral domain if it has no zero divisors.

Ex. In $\mathbb{Z}_{12}$, $\left. \begin{array}{l} 3 \cdot 4 = 0 \\ 2 \cdot 6 = 0 \\ 8 \cdot 9 = 0 \\ 6 \cdot 10 = 0 \end{array} \right\}$ zero divisors $= \{2, 3, 4, 6, 8, 9, 10\}$.
(Non-example). $\mathbb{Z}_{12}$ is NOT an integral domain.

Example: $\mathbb{Z}_{13} = \{0, 1, 2, \dots, 12\}$, every non zero element is a unit.

$\downarrow$
(Also, by property: a unit cannot be a zero divisor)

$\Rightarrow$ It has no zero divisors.

Therefore $\mathbb{Z}_{13}$ is an integral domain.

Property: (Cancellation in I.d.) Let a, b, c be elements of an integral domain R and suppose $a \neq 0$. If $ab = ac$ then $b = c$.

Remark. Important assumption $a \neq 0$. (otherwise
 $0 \cdot 4 = 0 \cdot 5$ in $\mathbb{Z}$
 $\nRightarrow 4 = 5$).

$$\mathbb{Z}_3[i] = \{0, 1, 2, i, 1+i, 2+i, 2i, 1+2i, 2+2i\}$$

$$= \{a+bi \mid a, b \in \mathbb{Z}_3\}, \quad i^2 = -1$$

is a field.

Inverse of $2+i \in \mathbb{Z}_3[i]$

$$= \exists a+bi \in \mathbb{Z}_3[i] \ni (2+i)(\underbrace{a+bi}) = 1$$

$$a=1, b=1.$$

Ex: $\mathbb{Q}(\sqrt{2}) = \{a+b\sqrt{2} \mid a, b \in \mathbb{Q}\}.$

Addition: $(2+15\sqrt{2}) + (7-8\sqrt{2}) = (2+7) + (15-8)\sqrt{2} = 9+7\sqrt{2}.$

Multiply: $(2+15\sqrt{2}) \cdot (7-8\sqrt{2})$

$$= 2 \cdot 7 + 2(-8\sqrt{2}) + 15\sqrt{2}(7) + 15\sqrt{2}(-8\sqrt{2})$$

$$= -226 + 89\sqrt{2}$$

Inverse: Inverse of $7+3\sqrt{2} \in \mathbb{Q}(\sqrt{2})$

$$= \frac{1}{7+3\sqrt{2}} = \frac{7-3\sqrt{2}}{(7+3\sqrt{2})(7-3\sqrt{2})} = \frac{7}{31} - \frac{3}{31}\sqrt{2}.$$

Inf $\mathbb{Q}(\sqrt{2})$ is a field.

Example: (Non example)

$\mathbb{Z}$: Integral domain, which is not a field.
only units are 1, -1.

Property: Every field is an integral domain.

Verification: $\mathbb{R}$ is a field

Claim: $\mathbb{R}$ is an integral domain.

$$\text{Let } \alpha, \beta \in \mathbb{R} \ni \alpha\beta = 0$$

$$\text{To show } \alpha = 0 \text{ or } \beta = 0$$

$$\text{Assume } \alpha \neq 0.$$

$$\text{Now, } \alpha\beta = \alpha \cdot 0 \Rightarrow \underbrace{\alpha^{-1}} \alpha\beta = \underbrace{\alpha^{-1}} \alpha \cdot 0$$

$$\Rightarrow \beta = 0 \quad (\text{since } \alpha \neq 0, \alpha^{-1} \text{ exists})$$

$$\alpha\alpha^{-1} \text{ is a unit.}$$

Inf $\mathbb{R}$ is an i.d.

Property: Any finite integral domain is a field.

Verification: Let R be a finite integral domain.

To show R is a field. That is, to show $0 \neq a \in R$ is a unit.

Let $f: R \rightarrow R$ defined by $f(x) = ax$.

f is injective:

$$f(x) = f(y)$$

$$\Rightarrow ax = ay$$

$$\Rightarrow a(x-y) = 0$$

$$\Rightarrow x-y = 0 \quad (\text{since } 0 \neq a \in R \text{ and } R \text{ is i.d. no zero divisors.})$$

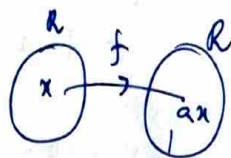
$$\Rightarrow x = y$$

f is surjective: Since R is finite, f is surjective.

Since $1 \in R$, $\exists b \in R \ni ab = 1$.

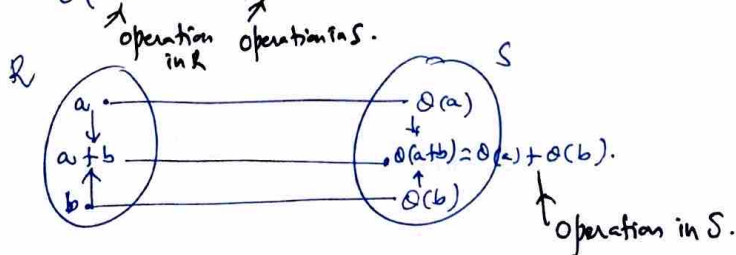
$\Rightarrow a$ is a unit.

Hence R is a field.



Def: (Ring homo). Let R and S be rings. A function

$\phi: R \rightarrow S$ is a ring homomorphism if $\phi(a+b) = \phi(a) + \phi(b)$, and $\phi(ab) = \phi(a) \cdot \phi(b)$ $\forall a, b \in R$.



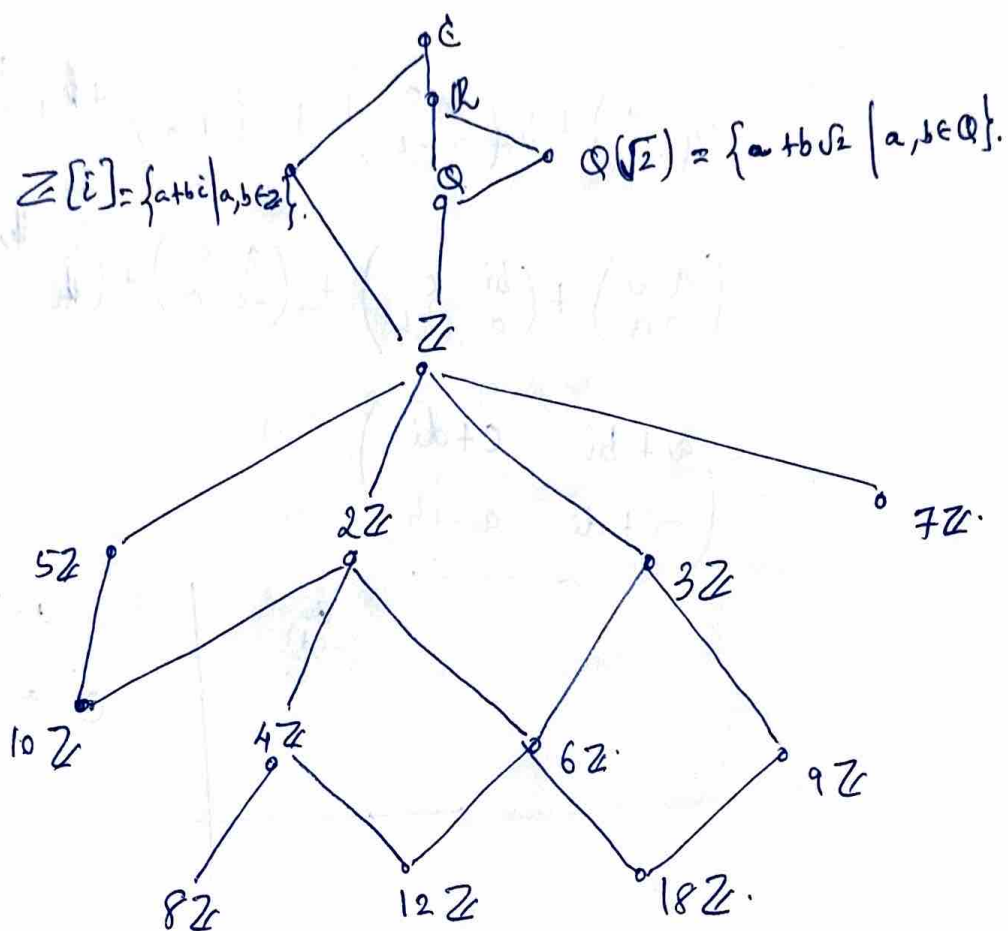
Ex: $\phi: \mathbb{Z} \rightarrow \mathbb{Z}_5$ by $\phi(a) = a \pmod{5} \forall a \in \mathbb{Z}$.

$$\phi(26+17) = \phi(43) = 43 \pmod{5} = 3 \pmod{5}$$

$$\begin{aligned} \phi(26) + \phi(17) &= 26 \pmod{5} + 17 \pmod{5} \\ &= 1 \pmod{5} + 2 \pmod{5} \\ &= 3 \pmod{5} \end{aligned}$$

$$\phi(26 \cdot 17) = \phi(442) = 2 \pmod{5}$$

$$\begin{aligned} \phi(26) \cdot \phi(17) &= 26 \pmod{5} \cdot 17 \pmod{5} \\ &= 1 \pmod{5} \cdot 2 \pmod{5} = 2 \pmod{5} \end{aligned}$$



Partial subring lattice diagram of $\mathbb{C}$.

Ex: $(\mathbb{Z}_{12}, +, \cdot)$

Units: $\{1, 5, 7, 11\}$.

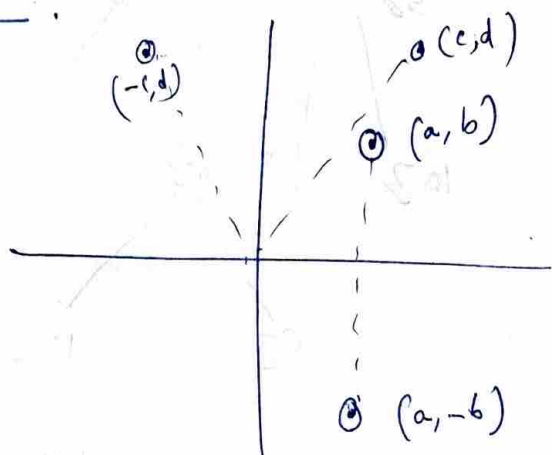
Zero divisors: $\{0, 2, 3, 4, 6, 8, 9, 10\}$.

Quaternions:

$$a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} + c \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

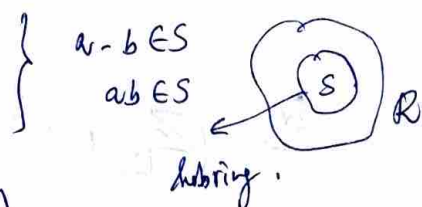
$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} + \begin{pmatrix} bi & 0 \\ 0 & -ib \end{pmatrix} + \begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix} + \begin{pmatrix} 0 & di \\ di & 0 \end{pmatrix}$$

$$= \begin{pmatrix} a+bi & c+di \\ -c+di & a-ib \end{pmatrix}.$$



Ex: $\{0\}, R$ are subrings.

1)



2) $\{0, 2, 4\}$ is a subring of $(\mathbb{Z}_6, +, \cdot)$.

3) for each $n \in \mathbb{Z}^+$, $n\mathbb{Z} = \{0, \pm n, \pm 2n, \pm 3n, \dots\}$ is a subring of $\mathbb{Z}$.

4) $\mathbb{Z}[i] = \{a+bi \mid a, b \in \mathbb{Z}\}$ is a subring of complex numbers $\mathbb{C}$.

Ex. 5) The set $\left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{Z} \right\}$ of diagonal matrices is a subring of the ring of all 2×2 -matrices over $\mathbb{Z}$.

Ex: $(\mathbb{Z}_n, +, \cdot)$ is a ring with 1.

1)

2) $(\mathbb{Z}_n, +, \cdot)$ is a ring.

$+$ = $+$: addition modulo n .

$\cdot$ = $\cdot$: multiplication modulo n .

3) $M_2(\mathbb{Z})$ = 2×2 matrices with integer entries.
is non-commutative ring with $1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

4) $M_2(\mathbb{R})$ = 2×2 matrices over real numbers.

5) A quaternion (in matrix form) is a 2×2 matrix of complex numbers of the form

$$\alpha = \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$$

$$\text{let } I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

$$\text{then } \alpha = aI + bi + cj + dk.$$

$$\text{let } H = \{ a + bi + cj + dk \mid a, b, c, d \in \mathbb{R} \}.$$

"Real Hamilton Quaternions"

$$\text{Prove: } \left. \begin{matrix} i^2 \\ j^2 \\ k^2 \end{matrix} \right\} = -1, \quad \left. \begin{matrix} ij \\ -ji \end{matrix} \right\} = k, \quad \left. \begin{matrix} jk \\ -kj \end{matrix} \right\} = i,$$

$$\left. \begin{matrix} ki \\ -ik \end{matrix} \right\} = j.$$

Cancellation property:

A ring has cancellation property if

$$(ab = ac \text{ or } ba = ca) \Rightarrow b = c$$

for any $a, b, c \in R, a \neq 0$.

Property: R has cancellation property $(\Rightarrow)$ it has no zero divisors of zero.

Verification: R has cancellation property (hypo.)

Claim: R has no non zero zero divisors

$$\text{let } ab = 0$$

$$\text{To show } a = 0 \text{ or } b = 0$$

$$\text{If } a = 0 \quad / \text{ done!}$$

$$\text{Otherwise } ab = 0 = a \cdot 0. \quad (\text{by def.})$$

$$\Rightarrow b = 0 \quad (\text{by cancellation property})$$

Converse: Suppose $ab = ac, a \neq 0$

$$\Rightarrow ab - ac = 0$$

$$\Rightarrow a(b - c) = 0 \quad (\text{dis. property})$$

$$\Rightarrow b - c = 0 \quad (\text{since } a \neq 0 \text{ and no zero divisors})$$

$$\Rightarrow b = c.$$

Integral domain: (Def.). Commu. ring with unity (1) having cancellation property (ie, no zero divisors).

Example: Consider the hom. $\phi: \mathbb{Z} \rightarrow \mathbb{Z}_5$ by

$$\phi(a) = a \pmod{5} \quad \forall a \in \mathbb{Z}.$$

$$\ker \phi = 5\mathbb{Z}. \quad (\text{veifind!})$$

Here: $5\mathbb{Z}$ is an additive subgroup of $\mathbb{Z}$.

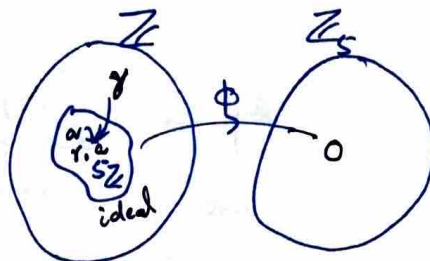
$$2) \quad r \in \mathbb{Z}, a \in 5\mathbb{Z}.$$

$$\Rightarrow r \cdot a \in 5\mathbb{Z}$$

$$r=7, a=10$$

$$r \cdot a = 7 \cdot 10 = 70$$

$$\in 5\mathbb{Z}.$$



Def: (Ideal of a ring) A subset A of a ring R is called an ideal of R if the following are true.

1) A is an additive subgroup of R

2) A satisfies: $r \in R, a \in A \Rightarrow r \cdot a \in A$. (and $a \cdot r \in A$)
(product absorption)

$5\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

R is an ideal of R } trivial ideals.

(0) " " " R }

Example: Consider $\phi: \mathbb{R}[x] \rightarrow \mathbb{R}$, where

$\mathbb{R}[x]$ is polynomials over reals, $\mathbb{R}$ is real numbers.

Define $\phi(f(x)) = f(2) \quad \forall f(x) \in \mathbb{R}[x]$.

Then $\ker \phi = \{ f(x) \in \mathbb{R}[x] \mid \phi(f(x)) = 0 \}$

For ex: let $p(x) = x^2 + x - 6$, $q(x) = x^3 - 7 \in \mathbb{R}[x]$.

Then $\phi(p(x)) = p(2)$ (by def.)

$$= 2^2 + 2 - 6$$

$$= 0$$

$$\Rightarrow \underline{p(x) \in \ker \phi}.$$

$$\begin{aligned}\text{But } \mathcal{O}(g(x)) &= \mathcal{O}(2) \\ &= 2^3 - 7 \\ &= 1 \neq 0.\end{aligned}$$

$$\Rightarrow \underline{g(x) \notin \ker \mathcal{O}}.$$

In general, $f(x) \in \ker \mathcal{O}$ when $\mathcal{O}(f(x)) = 0$
i.e., $f(2) = 0$.

By factor property (theorem), there are polynomials that have $x-2$ as a factor, such that

$$p(x) = x^2 + x - 6 = (x-2)(x+3)$$

$$\text{Thus } \ker \mathcal{O} = \{ (x-2) \cdot g(x) \mid g(x) \in \mathbb{R}[x] \}.$$

Next verify: $\ker \mathcal{O}$ is an ideal of $\mathbb{R}[x]$.

By property —, we have $\ker \mathcal{O}$ is an additive subgroup of $\mathbb{R}[x]$.

Now let $f(x) \in \mathbb{R}[x]$ and $p(x) \in \ker \mathcal{O}$.

To show $f(x) \cdot p(x) \in \ker \mathcal{O}$. (product absorption property)

$$p(x) \in \ker \mathcal{O} \Rightarrow p(x) = (x-2) \cdot g(x) \text{ for some } g(x) \in \mathbb{R}[x].$$

$$\begin{aligned}\text{now } \mathcal{O}(f(x) \cdot p(x)) &= \mathcal{O}(f(x)) \cdot \mathcal{O}(p(x)) \\ &= f(2) \cdot p(2) \\ &= f(2) \cdot \underbrace{(2-2) \cdot g(2)}_0 \\ &= 0\end{aligned}$$

$$\Rightarrow f(x) \cdot p(x) \in \ker \mathcal{O}.$$

Thus $\ker \mathcal{O}$ is an ideal.

Property: (Kernel is an ideal): Let $\mathcal{O}: R \rightarrow S$ be a ring homo. with $K = \ker \mathcal{O}$. Then K is an ideal of R (domain).

Verification: clearly K is a additive subgp of R .

To see product absorption property:

Let $r \in R$ and $k \in K$.

$$\begin{aligned} \text{Now } \phi(r \cdot k) &= \phi(r) \cdot \phi(k) \\ &= \phi(r) \cdot 0_s \quad (\text{since } k \in \text{Ker } \phi, \phi(k) = 0_s) \\ &= 0_s. \end{aligned}$$

Therefore $r \cdot k \in K$. (similar way, $k \cdot r \in K$).

Thus K is an ideal of R .

Def. (principal ideal): Let R be a commu. ring, $a \in R$ (fixed elt).

Then the set $\langle a \rangle = \{a \cdot r \mid r \in R\}$ is called the principal ideal generated by a . (ie, set of all multiples of ' a ').

(very similar to cyclic subgp generated by ' a ' in gpt.)

Ex. $5 \in \mathbb{Z}$, $\langle 5 \rangle = \{5 \cdot r \mid r \in \mathbb{Z}\} = 5\mathbb{Z}$, an ideal of $\mathbb{Z}$

$5 \in \mathbb{Z}_{10}$, $\langle 5 \rangle = \{5 \cdot r \mid r \in \mathbb{Z}_{10}\} = \{0, 5\}$, an ideal of $\mathbb{Z}_{10}$

$x-2 \in \mathbb{R}[x]$, $\langle x-2 \rangle = \{(x-2) \cdot g(x) \mid g(x) \in \mathbb{R}[x]\}$, an ideal of $\mathbb{R}[x]$.

$1 \in R$ (ring), $\langle 1 \rangle = \{1 \cdot r \mid r \in R\} = R$.
 $0 \in R$ ("), $\langle 0 \rangle = \{0 \cdot r \mid r \in R\} = \{0\}$ } Trivial ideals.

Property: Let R : comm. ring, $a \in R$ (fixed). Then

$\langle a \rangle = \{a \cdot r \mid r \in R\}$ is an ideal of R .

From a quotient group to a quotient ring:

$\mathbb{Z}$: additive gp., $5\mathbb{Z} \leq \mathbb{Z}$.

Coset of $5\mathbb{Z}$: $a+5\mathbb{Z}$, $a \in \mathbb{Z}$.

Observe that: $4378 + 5\mathbb{Z} = 3 + 5\mathbb{Z}$ in $\mathbb{Z}/5\mathbb{Z}$
but $4378 \neq 3$ in $\mathbb{Z}$.

$$\begin{cases} a+5\mathbb{Z} = b+5\mathbb{Z} \iff a-b \in 5\mathbb{Z} \\ a+5\mathbb{Z} = 0+5\mathbb{Z} \iff a \in 5\mathbb{Z} \text{ (by taking } b=0 \text{ above).} \end{cases}$$

$$\mathbb{Z}/5\mathbb{Z} = \{0+5\mathbb{Z}, 1+5\mathbb{Z}, 2+5\mathbb{Z}, 3+5\mathbb{Z}, 4+5\mathbb{Z}\}.$$

$$(a+5\mathbb{Z}) + (b+5\mathbb{Z}) = (a+b) + 5\mathbb{Z}.$$

$$(3+5\mathbb{Z}) + (4+5\mathbb{Z}) = 7+5\mathbb{Z} = 2+5\mathbb{Z}.$$

$$\left\{ \begin{array}{l} \text{Remember} \\ \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}_5 \end{array} \right.$$

$$a+5\mathbb{Z} \leftrightarrow a(\text{mod } 5).$$

Define multiplication in $\mathbb{Z}/5\mathbb{Z}$.

$$(a+5\mathbb{Z}) \cdot (b+5\mathbb{Z}) = a \cdot b + 5\mathbb{Z}.$$

$$\underbrace{(2+5\mathbb{Z}) \cdot (3+5\mathbb{Z})}_{\text{inverses of each other}} = 6+5\mathbb{Z} = 1+5\mathbb{Z}.$$

Def: (Quotient ring): Let R be a comm. ring, A is an ideal of R .

The set of cosets $R/A = \{r+A \mid r \in R\}$ is a quotient ring under the operations:

$$(r+A) + (s+A) = (r+s) + A$$

$$(r+A) \cdot (s+A) = (r \cdot s) + A.$$

Note: Suppose $2+5\mathbb{Z} = 22+5\mathbb{Z}$ and $3+5\mathbb{Z} = 58+5\mathbb{Z}$.

$$\text{Then } (22+5\mathbb{Z}) \cdot (58+5\mathbb{Z}) = (2+5\mathbb{Z}) \cdot (3+5\mathbb{Z}).$$

Verification: $(22+5\mathbb{Z}) \cdot (58+5\mathbb{Z})$

$$= ((2+5 \cdot 4) + 5\mathbb{Z}) \cdot ((3+5 \cdot 11) + 5\mathbb{Z})$$

$$= (2+5 \cdot 4) \cdot (3+5 \cdot 11) + 5\mathbb{Z} \text{ (def.)}$$

$$= (2 \cdot 3 + 2(5 \cdot 11) + (5 \cdot 4) \cdot 3 + (5 \cdot 4)(5 \cdot 11)) + 5\mathbb{Z}.$$

By division algorithm

$$22 = 2 + 5 \cdot 4$$

$$58 = 3 + 5 \cdot 11.$$

$$= 2 \cdot 3 + 5\mathbb{Z}.$$

$$= (2+5\mathbb{Z}) \cdot (3+5\mathbb{Z}).$$

Property: (Coset multiplication in $\mathbb{Z}/5\mathbb{Z}$ is well defined):

$$\text{Suppose } a+5\mathbb{Z} = c+5\mathbb{Z} \text{ and}$$

$$b+5\mathbb{Z} = d+5\mathbb{Z}.$$

$$\text{Then } (a+5\mathbb{Z}) \cdot (b+5\mathbb{Z}) = (c+5\mathbb{Z}) \cdot (d+5\mathbb{Z}).$$

Proof:

$$a+5\mathbb{Z} = c+5\mathbb{Z}$$

$$\Rightarrow a-c \in 5\mathbb{Z}$$

$$\Rightarrow a = c+x \text{ for some } x \in 5\mathbb{Z}.$$

$$\text{Similarly, } b+5\mathbb{Z} = d+5\mathbb{Z}$$

$$\Rightarrow b = d+y \text{ for some } y \in 5\mathbb{Z}.$$

$$\text{Now } (a+5\mathbb{Z}) \cdot (b+5\mathbb{Z}) = ((c+x)+5\mathbb{Z}) \cdot ((d+y)+5\mathbb{Z})$$

$$= (c+x) \cdot (d+y) + 5\mathbb{Z}$$

$$= c \cdot d + \underbrace{(c \cdot y + x \cdot d + x \cdot y)}_{\in 5\mathbb{Z} \text{ as it is an ideal of } \mathbb{Z}} + 5\mathbb{Z}.$$

$\in 5\mathbb{Z}$ as it is an ideal of $\mathbb{Z}$.

$$= c \cdot d + 5\mathbb{Z}$$

$$= (c+5\mathbb{Z})(d+5\mathbb{Z}).$$

Ex. Consider the poly. x^2+1 . (Say $f(x)$)

$$\text{In } \mathbb{Z}_5[x]: \quad x^2+1$$

$$= (x+2) \cdot (x+3) \quad (\text{deg. of each are "less than" the given poly.})$$

$$= x^2 + \underline{5}x + 6$$

$$= x^2 + \underline{0}x + 1 \quad (\text{since } 5 \equiv 0 \pmod{5})$$

$$= x^2 + 1$$

Therefore $f(x)$ is factorable in $\mathbb{Z}_5[x]$;

and "unfactorable" in $\mathbb{R}[x]$.

However, both $\mathbb{R}[x]$ and $\mathbb{Z}_5[x]$ are polynomial rings.

Remark: (It is not a legitimate factorization).

$$x^2+1 = 3 \cdot \left(\frac{1}{3}x^2 + \frac{1}{3} \right)$$

Reason: When we factor a polynomial, we must write it as a product of two "smaller" polynomials.

Here, $\frac{1}{3}x^2 + \frac{1}{3}$ is NOT ^{deg.} smaller than x^2+1 .

Def. (Factorable/unfactorable polynomials.) Let F be a field. Suppose $f(x) \in F[x]$ with $\deg f(x) \geq 1$ (ie., $f(x)$ is not a constant poly.).

We say that $f(x)$ is factorable in $F[x]$ when

$$f(x) = p(x) \cdot q(x) \quad \text{for some } p(x), q(x) \in F[x] \\ \deg p(x), q(x) < \deg f(x).$$

Otherwise, $f(x)$ is unfactorable in $F[x]$.

Ex. let $f(x) = x^2-2$ in $\mathbb{R}[x]$ is factorable.

$$\text{Now } x^2-2 = (x+\sqrt{2}) \cdot (x-\sqrt{2}) \quad \text{where } \begin{matrix} x+\sqrt{2} \in \mathbb{R}[x] \\ x-\sqrt{2} \in \mathbb{R}[x]. \end{matrix}$$

$$\begin{array}{ccc} \swarrow & \uparrow & \uparrow \\ \deg 2 & \deg 1 & \deg 1 \end{array}$$

Ex: Let $f(x) = x^2 + 1$ in $\mathbb{R}[x]$.

Claim: $f(x)$ is unfactorable.

On a contrary: $f(x) = p(x) \cdot q(x)$ where
 $\deg p(x) < \deg f(x)$
 $\deg q(x) < \deg f(x)$

Since $\deg f(x) = 2$, we must have

$\deg p(x) = 1 = \deg q(x)$, linear poly;

Say: $\begin{cases} p(x) = x + \alpha \\ q(x) = x + \beta \end{cases}$ where $\alpha, \beta \in \mathbb{R}$.

$$\text{Now } x^2 + 1 = (x + \alpha) \cdot (x + \beta)$$

$$= x^2 + (\alpha + \beta)x + \alpha\beta.$$

Equating coeff: $\alpha + \beta = 0$, $\alpha\beta = 1$.

This system has no solution in $\mathbb{R}$.

∴ $f(x)$ is unfactorable in $\mathbb{R}[x]$.

Ex: Let $f(x) = x^2 + x + 1$ in $\mathbb{Z}_2[x]$.

Verify: $f(x)$ is unfactorable in $\mathbb{Z}_2[x]$.

Contrary: $x^2 + x + 1 = p(x) \cdot q(x)$, where $p(x), q(x)$
have $\deg. 1$.

The following possibilities:

The $\deg 1$ polys in $\mathbb{Z}_2[x]$ are 'x' and 'x+1'

(i) $p(x) = x, q(x) = x+1$ (or vice-versa) $\Rightarrow p(x) \cdot q(x) = x^2 + 1$

(ii) $p(x) = x, q(x) = x \Rightarrow p(x) \cdot q(x) = x^2$

(iii) $p(x) = x+1, q(x) = x+1 \Rightarrow p(x) \cdot q(x) = x^2 + 2x + 1$
 $= x^2 + 1$ (since $2 \equiv 0$ mod 2)

In all cases: $f(x) \neq p(x) \cdot q(x)$

∴ $f(x)$ is unfactorable in $\mathbb{Z}_2[x]$.

Theorem: (property) Let F be a field. Suppose $f(x) \in F[x]$ with $\deg f(x) = 1$. Then $f(x)$ is unfactorable in $F[x]$.

Verification: Suppose $f(x) = p(x) \cdot q(x)$ for some $p(x), q(x) \in F[x]$. Since $\deg f(x) = \deg p(x) + \deg q(x)$ and $\deg f(x) = 1$, it follows that either $\deg p(x) = 0$ or $\deg q(x) = 0$. Then by property... we get that $f(x)$ is unfactorable.

Ex. $f(x) = x^2 + 1$ in $\mathbb{Z}_5[x]$.

$$f(3) = 3^2 + 1 = 10 = 0 \text{ (in } \mathbb{Z}_5)$$

$\Rightarrow 3 \in \mathbb{Z}_5$ is a root of $f(x)$.

However, $f(4) = 4^2 + 1 = 17 \neq 0$ in $\mathbb{Z}_5$ (4 is not a root of $f(x)$ in $\mathbb{Z}_5$)

Def. (Root of a poly.).

Let F be a field, let $f(x) \in F[x]$. We say that an element $\alpha \in F$ is a root of the poly. $f(x)$ if $f(\alpha) = 0$.

Ex. Let $f(x) = 4x + 5$ in $\mathbb{R}[x]$.

$$f\left(-\frac{5}{4}\right) = 4 \cdot \left(-\frac{5}{4}\right) + 5 = -5 + 5 = 0.$$

$\Rightarrow 4 \cdot \alpha + 5 = 0$ for α .

Ex. $f(x) = 4x + 5$ in $\mathbb{Z}_{11}[x]$.

To find ' α ', solve: $4 \cdot \alpha + 5 = 0$

$$\Rightarrow \alpha = 4^{-1} \cdot (-5).$$

In $\mathbb{Z}_{11}$, we have $4^{-1} = 3$ (since $4 \cdot 3 = 12 \equiv 1 \pmod{11}$)

and $-5 = 6$ (since $5 + 6 = 11 = 0$ in $\mathbb{Z}_{11}$)

$$\text{Thus } \alpha = 3 \cdot (6) = 18 \equiv 7 \pmod{11}$$

Hence, $f(7) = 4 \cdot 7 + 5 = 33 \equiv 0 \pmod{11}$
'7' is a root of $f(x)$ in $\mathbb{Z}_{11}[x]$.

Property: (Theorem): Let F be a field. Suppose $f(x) \in F[x]$ with $\deg f(x) \geq 1$. Then $f(x)$ is unfactorable if and only if $f(x)$ satisfies the following property:

If $f(x) = p(x) \cdot q(x)$, then $\deg p(x) = 0$ or $\deg q(x) = 0$.

Proof: Suppose $f(x)$ is unfactorable.

Assume that $f(x) = p(x) \cdot q(x)$, where $p(x), q(x) \in F[x]$.

Now by property (refer--),

$$\deg f(x) = \deg p(x) + \deg q(x) \quad \text{①} \quad \left(\begin{array}{l} \text{since } F[x] \text{ is a} \\ \text{field} \Rightarrow \text{i-domain} \end{array} \right)$$

$$\Rightarrow \deg p(x), \deg q(x) \leq \deg f(x)$$

Here, $\deg p(x), \deg q(x)$ both cannot be strictly less than $\deg f(x)$ (by hypo.)

Therefore either $\deg p(x) = \deg f(x)$
or $\deg q(x) = \deg f(x)$

If $\deg p(x) = \deg f(x)$, then $\deg q(x) = 0$ (by ①)

" $q(x) =$ " " $\deg p(x) = 0$ "

Converse: Try!

Questions:

1) Consider $f(x) = 2x - 7$ in $\mathbb{R}[x]$. (unfactorable)

(since not possible $f(x)$ into smaller that are not constants).

2) $f(x) = 4x + 2$ in $\mathbb{Z}_7[x]$. is unfactorable in $\mathbb{Z}_7[x]$,

Hint: $ax + b \in F[x]$, $a \neq 0$ is unfactorable in $F[x]$.

3)

Polynomial Ring.

Let $\mathbb{Z}[x]$: set of polynomials with coefficients from $\mathbb{Z}$.

Let $f(x) = 3x^4 - 7x^2 + 4$, $g(x) = 4x^2 + 1$.

$$\begin{aligned} f(x) + g(x) &= (3x^4 - 7x^2 + 4) + (4x^2 + 1) \\ &= 3x^4 + (-7x^2 + 4x^2) + (4 + 1) \\ &= 3x^4 - 3x^2 + 5. \end{aligned}$$

$$\begin{aligned} f(x) \cdot g(x) &= 3x^4(4x^2 + 1) - 7x^2(4x^2 + 1) + 4(4x^2 + 1) \\ &= 12x^6 - 28x^4 + 9x^2 + 4. \end{aligned}$$

Compute $f(x) \cdot g(x)$: (Area model)

$$\leftarrow 4x^2 + 1 \rightarrow g(x)$$

	$3x^4$	$12x^6$ Area	$3x^4$ Area
	$+$		
	$-7x^2$	$-28x^4$ Area	$-7x^2$ Area
	$+$		
	4	$16x^2$ Area	4 Area
$f(x)$			

Product of $f(x)$ and $g(x)$ is the "total area in the squares".

$$-f(x) = -(f(x)) = -3x^4 + 7x^2 - 4.$$

$$f(x) + (-f(x)) = 0.$$

$\mathbb{Z}[x]$ is a ring. Also it's a commutative ring.

Ex: Ring of integers $\mathbb{Z}$ is a subring of $\mathbb{Z}[x]$.

$0, 1, -3$ and 6 are constant polynomials.

Def: (polynomial ring). Let R be a commutative ring. Then $R[x]$ is called a polynomial ring, is the set of all polynomials with coefficients from R .

An element $f(x) \in R[x]$ is of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0; \quad a_i \in R.$$

n : non-negative.

Example: R is a subring of $R[x]$.

Ex: $\mathbb{Z}_5[x]$: polynomial ring with coefficients from $\mathbb{Z}_5$.

$\mathbb{Z}_5$ can be viewed as subring of $\mathbb{Z}_5[x]$ where

$0, 1, 2, 3, 4$ are constant polynomials.

In $\mathbb{Z}_5[x]$, $7x = 2x$ but $x^7 \neq x^2$. / Important to note.

Def: (deg. of poly.) let $0 \neq f(x) \in R[x]$ where R is a comm. ring.
 $\deg f(x) =$ highest exponent of $f(x)$.

Ex: let $f(x) \in R[x]$ be a linear polynomial. i.e., $f(x) = a_1x + a_0$
where $a_1 \neq 0$. Then $f(x) = a_1x^1 + a_0x^0$.
 $\deg f(x) = 1$.

Ex: let $f(x) = 7 \in R[x]$ is a constant polynomial.
i.e., $f(x) = 7x^0$.

Def: A poly. is said to be monic if its leading coeff is 1.

Ex: $f(x) = x^6 + 7x^3 - 2x + 3$ is monic.

Ex: Consider $f(x) = 3x^{15} + 4x^3 + 2$
 $g(x) = 6x^8 + 5x + 3$.

In $\mathbb{Z}[x]$, $\deg f(x) = 15$, $\deg g(x) = 8$
 $f(x) \cdot g(x) = (3x^{15})(6x^8) + \text{Lower deg. terms.}$
 $= (3 \cdot 6)x^{15+8} + \text{ " " }$
 $= 18x^{23} + \text{ " " }$

Therefore $\deg f(x) \cdot g(x) = 23$.

In $\mathbb{Z}_7[x]$ $f(x) \cdot g(x) = 4x^{23}$ (Since $18 \equiv 4 \pmod{7}$)
Therefore $\deg f(x) \cdot g(x) = 23$.

In $\mathbb{Z}_9[x]$. $\deg f(x) = 15, \deg g(x) = 8.$

However, compute...

$$\begin{aligned}
 f(x) \cdot g(x) &= (3x^{15} + 4x^3 + 2) \cdot (6x^8 + 5x + 3) \\
 &= (3 \cdot 6)x^{15+8} + (3 \cdot 5)x^{15+1} + (\text{lower deg terms}) \\
 &= 18x^{23} + 15x^{16} + (\text{lower deg terms}) \\
 &= \underline{0}x^{23} + \underline{6}x^{16} + (\text{lower deg terms}) \\
 &\quad \left(\begin{array}{l} \text{Since } 18 \equiv 0 \pmod{9} \\ 15 \equiv 6 \pmod{9} \end{array} \right) \\
 &= 6x^{16} + (\text{lower deg terms})
 \end{aligned}$$

Therefore $\deg f(x) \cdot g(x) = 16$, but NOT 23.

The above example leads to the following theorem.

Theorem: (property): (Degree of a product)

Suppose R is an integral domain. let $f(x), g(x) \in R[x]$ with $f(x) \neq 0 \neq g(x)$. Then

$$\deg(f(x) \cdot g(x)) = \deg f(x) + \deg g(x).$$

Proof. (verification): Since $f(x)$ and $g(x)$ are non zero polynomials,

$$\text{assume that } f(x) = a_m x^m + a_{m-1} x^{m-1} + \dots + a_1 x + a_0,$$

$$g(x) = b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0,$$

where $a_m \neq 0 \neq b_n$.

now $\deg f(x) = m, \deg g(x) = n$.

$$\begin{aligned}
 f(x) \cdot g(x) &= (a_m x^m) \cdot (b_n x^n) + (\text{lower deg terms}) \\
 &= a_m \cdot b_n x^{m+n} + (\text{lower deg terms})
 \end{aligned}$$

Since R is an integral domain, (hypo.),

$$a_m \cdot b_n \neq 0.$$

$$\text{Therefore } \deg(f(x) \cdot g(x)) = m+n.$$

$$= \deg f(x) + \deg g(x).$$

Ex: (Non-example).

Consider the poly ring $\mathbb{Z}_9[x]$.

Note that $\mathbb{Z}_9$ is not an integral domain, since
 $3 \cdot 6 = 0$ in $\mathbb{Z}_9$. (3 and 6 are zero divisors).

$$(3x+1) \cdot (6x+1) = 18x^2 + 9x + 1$$

$$= \underline{0}x^2 + 0x + 1$$

$$= 1 \text{ (unit)}$$

Therefore the non-constant polynomials $(3x+1)$ and $(6x+1)$ are units in $\mathbb{Z}_9[x]$.

Ex: (non-example) Consider $\mathbb{Z}_6[x]$, where $\mathbb{Z}_6$ is not an integral domain.

Consider non-zero polynomials $4x+2$ and $3x$ in $\mathbb{Z}_6[x]$.

$$(4x+2) \cdot 3x = 12x^2 + 6x$$

$$\underbrace{\quad \quad \quad}_{\nearrow} \underbrace{\quad \quad \quad}_{\nearrow} = 0x^2 + 0x = 0.$$

zero divisors in $\mathbb{Z}_6[x]$.

Therefore $\mathbb{Z}_6[x]$ is not an integral domain.

Ring Isomorphism:

Let R and S be rings. A function $f: R \rightarrow S$ is a ring homomorphism if f is a bijection (i.e., 1-1 and onto) and f preserves both addition and multiplication. (isomorphism).

Example: Consider the function

$$\phi: \mathbb{Z}_2[x] \rightarrow \mathbb{Z}_2[x] \text{ by}$$

$$\phi(f(x)) = (f(x))^2, \quad \forall f(x) \in \mathbb{Z}_2[x].$$

Take $f(x) = x^2 + 1 \in \mathbb{Z}_2[x]$. Then

$$\phi(f(x)) = (x^2 + 1)^2$$

$$= (x^2 + 1) \cdot (x^2 + 1)$$

$$= x^4 + 2x^2 + 1$$

$$= x^4 + 0x^2 + 1. \quad (\text{since } 2 \equiv 0 \pmod{2})$$

$$= x^4 + 1.$$

For any $f(x), g(x) \in \mathbb{Z}_2[x]$,

$$\phi(f(x) + g(x)) = (f(x) + g(x))^2$$

$$= f(x)^2 + 2 \cdot f(x)g(x) + g(x)^2$$

$$= f(x)^2 + g(x)^2 \quad (\text{since } 2 \equiv 0 \pmod{2})$$

$$= \phi(f(x)) + \phi(g(x))$$

Similarly,

$$\phi(f(x) \cdot g(x)) = (f(x) \cdot g(x))^2$$

$$= f(x) \cdot g(x) \cdot f(x) \cdot g(x)$$

$$= \underbrace{f(x) \cdot f(x)}_{\leftrightarrow} \cdot g(x) \cdot g(x) \quad (\text{since } \mathbb{Z}_2[x] \text{ is comm.})$$

$$= f(x)^2 \cdot g(x)^2$$

$$= \phi(f(x)) \cdot \phi(g(x)).$$

Thus ϕ is a ring homo.

Ex: (Non-example). Consider the det function $\delta: M(\mathbb{Z}_{10}) \rightarrow \mathbb{Z}_{10}$
where $\delta(\alpha) = \det \alpha \quad \forall \alpha \in M(\mathbb{Z}_{10})$.

For $\alpha, \beta \in M(\mathbb{Z}_{10})$,

$$\begin{aligned}\delta(\alpha \cdot \beta) &= \det(\alpha \cdot \beta) \\ &= \det \alpha \cdot \det \beta \\ &= \delta(\alpha) \cdot \delta(\beta).\end{aligned}$$

} upto this gp
have. (multiplicatively)

δ fails to preserve addition:

Take $\alpha = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\beta = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \in M(\mathbb{Z}_{10})$.

$$\begin{aligned}\delta(\alpha + \beta) &= \delta\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}\right) \\ &= \delta\left(\begin{bmatrix} 6 & 8 \\ 0 & 2 \end{bmatrix}\right) \quad (\text{in } \mathbb{Z}_{10}) \\ &= \det \begin{pmatrix} 6 & 8 \\ 0 & 2 \end{pmatrix} \\ &= 12 \equiv 2 \pmod{10}\end{aligned}$$

$$\begin{aligned}\delta(\alpha) + \delta(\beta) &= \det \alpha + \det \beta \\ &= 1 + 1 \\ &= 2 \equiv 2 \pmod{10}\end{aligned}$$

∴ $\delta(\alpha + \beta) \neq \delta(\alpha) + \delta(\beta)$.

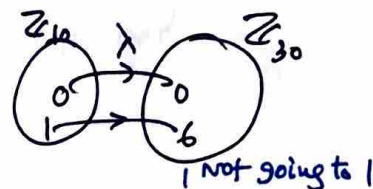
Properties: $\phi: R \rightarrow S$ ring hom. additive gps.

1) $\phi(0_R) = 0_S$.

Ex: Consider $\lambda: \mathbb{Z}_{10} \rightarrow \mathbb{Z}_{30}$ where $\lambda(a) = 6a, \forall a \in \mathbb{Z}_{10}$.

$$\lambda(0) = 6 \cdot 0 = 0.$$

↑ in $\mathbb{Z}_{10}$ ↑ $\mathbb{Z}_{30}$.



2) $\phi(1_R)$ need not be equal to 1_S .

$$\lambda(1) = 6 \cdot 1 = 6 (\neq 1 \text{ in } \mathbb{Z}_{30}).$$

∴ $\lambda(1) \neq 1$.

Def: (kernel): Let $\phi: R \rightarrow S$ be a ring homomorphism.

The kernel of ϕ is the set $\ker \phi = \{r \in R \mid \phi(r) = 0_S\}$.

Ex: Consider $\phi: \mathbb{Z} \rightarrow \mathbb{Z}_5$ by $\phi(a) = a \pmod{5}$.

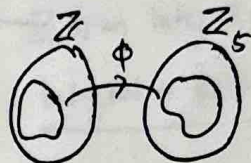
Then ϕ is a hmo.

$$\forall a \in \mathbb{Z}.$$

$$\text{Ker } \phi = \{x \in \mathbb{Z} \mid \phi(x) = 0 \pmod{5}\}.$$

$$= \{\text{all multiples of } 5\}.$$

$$= 5\mathbb{Z}.$$



$$\text{Im } \phi = \mathbb{Z}_5 = \{x \in \mathbb{Z}_5 \mid x \in \mathbb{Z}\}.$$

Observation: $\text{Ker } \phi$ is an additive subgroup of $\mathbb{Z}$

$\text{Im } \phi$ " " " $\mathbb{Z}_5$.

Ex: Consider the ring hmo. $\lambda: \mathbb{Z}_{10} \rightarrow \mathbb{Z}_{30}$ defined by

$$\left. \begin{array}{l} \text{Compute} \\ \text{Ker } \lambda \\ \text{Im } \lambda \end{array} \right\} \begin{array}{l} \lambda(a) = 6a \quad \forall a \in \mathbb{Z}_{10}. \\ \lambda(0) = 6 \cdot 0 = 0. \\ \lambda(5) = 6 \cdot 5 = 0 \quad (\text{since } 30 \equiv 0 \pmod{30}) \end{array}$$

and $\lambda(a) \neq 0$ for all $a \notin \{0, 5\}$.

$$\text{Therefore } \text{Ker } \lambda = \{0, 5\}.$$

Compute $\lambda(a)$: $\lambda(0) = 0$

$$\lambda(1) = 6$$

$$\lambda(2) = 12$$

$$\lambda(3) = 18$$

$$\lambda(4) = 24.$$

$$\text{Therefore } \text{Im } \lambda = \{0, 6, 12, 18, 24\} \leq \mathbb{Z}_{30}.$$

Division algorithm in $\mathbb{Z}$: Let $a, b \in \mathbb{Z}$, $b > 0$. Then $\exists q, r \in \mathbb{Z}$

$$\text{such that } a = b \cdot q + r, \quad 0 \leq r < b.$$

Ex: (polynomial): Consider $f(x) = 5x^4 + x^3 - 3x^2 + 4x - 3$ and

$$g(x) = x^2 + 1 \quad \text{in } \mathbb{R}[x], \quad \mathbb{R} \text{ is a field. (why!)}$$

Use division algorithm to find quotient and the remainder.

$$\begin{array}{r}
 \boxed{5x^2 + x - 8} \leftarrow \text{quotient} \\
 \hline
 \underbrace{x^2 + 1}_{g(x)} \overline{) 5x^4 + x^3 - 3x^2 + 4x - 3} \leftarrow f(x) \\
 \underline{5x^4 \quad + 5x^2} \\
 x^3 - 8x^2 + 4x - 3 \\
 \underline{x^3 \quad + x} \\
 - 8x^2 + 3x - 3 \\
 \underline{- 8x^2 \quad - 8} \\
 \boxed{3x + 5} \leftarrow \text{remainder}
 \end{array}$$

Therefore $f(x) = (x^2 + 1) \cdot q(x) + (3x + 5)$,
 $q(x) = 5x^2 + x - 8$, $\deg(r(x)) < \deg(g(x))$
 $ = 1 \qquad = 2$.

Ex: Consider $f(x) = 4x^3 + 5x^2 + 2$,

$g(x) = 3x^2 + 5$ in $\mathbb{Z}_7[x]$, where $\mathbb{Z}_7$ is a field.

$$\begin{array}{r}
 \boxed{6x + 4} \leftarrow \text{quotient} \\
 \hline
 \underbrace{3x^2 + 5}_{g(x)} \overline{) 4x^3 + 5x^2 + 2} \leftarrow f(x) \\
 \underline{4x^3 \quad + 2x} \\
 5x^2 + 5x + 2 \quad \text{(here } -2x = 5x \text{)} \\
 \underline{- 5x^2 \quad + 6} \quad \text{(calculated as mod 7)} \\
 \boxed{5x + 3} \leftarrow \text{remainder} \quad (-4 \equiv 3 \pmod{7}).
 \end{array}$$

Therefore.

$$f(x) = (3x^2 + 5) q(x) + (5x + 3)$$

$\uparrow$ $$
 $6x + 4$ $r(x)$

$\deg r(x) = 1 < 2 = \deg g(x)$

Division algorithm (for poly.) Let F be a field. Suppose $f(x), g(x) \in F[x]$,
 $g(x) \neq 0$. Then $\exists \ q(x), r(x) \in F[x]$ such that $f(x) = g(x) \cdot q(x) + r(x)$,
 $r(x) = 0$ or $\deg r(x) < \deg g(x)$

Remark: Coefficients from field is must.

Example: (Non-example):

$$\text{Let } \begin{cases} f(x) = x^3 \\ g(x) = 2x \end{cases} \text{ in } \mathbb{Z}[x]$$

Note:

$\mathbb{Z}$ is an integral domain,
but not a field.

By division algorithm, (if exists here!)

$\exists q(x), r(x) \in \mathbb{Z}[x]$ such that

$$x^3 = \underbrace{2x}_{g(x)} \cdot q(x) + r(x), \quad \deg(r(x)) = 0 \text{ or } \deg r(x) < \deg g(x).$$

Since $\deg g(x) = 1$, we know $r(x)$ is a constant polynomial, possibly '0'.

Let $r(x) = m$ for some $m \in \mathbb{Z}$ (constant polynomial)

For the eqn. $x^3 = 2x \cdot q(x) + r(x)$ to be true, $q(x)$ must be a quadratic i.e., $q(x) = ax^2 + bx + c$ for some integers a, b, c .

$$\begin{aligned} \text{Thus we have } x^3 &= 2x(ax^2 + bx + c) + m \\ &= 2ax^3 + 2bx^2 + 2cx + m. \end{aligned}$$

Equating coefficients of x^3 :

$$\boxed{1 = 2a, \text{ cannot occur in } \mathbb{Z}.}$$

↑
This equation has no integer solution.

Ex: Let $f(x) = 3x^4 - 5x^3 - 8 \in \mathbb{R}[x]$
↙ field.

$$f(2) = 0 \text{ (verify!)}$$

$\Rightarrow x-2$ is a factor of $f(x)$.

$$\Rightarrow \exists v(x) \in \mathbb{R}[x] \ni f(x) = (x-2)v(x).$$

Factor property: Let F be a field, $a \in F$, $f(x) \in F[x]$.

Then $f(a) = 0 \iff (x-a) \mid f(x)$. (Map proof!)

Ex: Consider $f(x) = 5x^{672} + 2x^{359} + 4x^{101} + x^{77} + 3x^{23} + 6$ in $\mathbb{Z}_7[x]$.

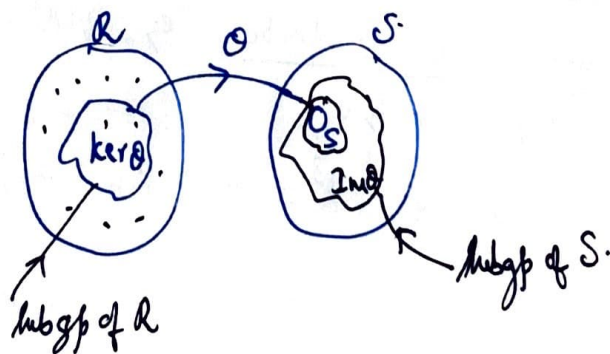
$$f(1) = 5 + 2 + 4 + 1 + 3 + 6 = 21 = 0 \pmod{7}.$$

$\Rightarrow f(1) = 0$ i.e., $(x-1)$ is a factor of $f(x)$

But $f(-1) = 1$ (verify!) $\Rightarrow x - (-1) = x+1$ is not a factor of $f(x)$

Property: Let $\phi: R \rightarrow S$ be a ring homomorphism. Then

- 1) $\ker \phi$ is an additive subgroup of the domain R (refer gp. theory!).
- 2) $\text{Im } \phi$ is an additive subgroup of the co-domain S .



Ex. Consider the ring

$$\mathbb{Z}_5[i] = \{a + bi \mid a, b \in \mathbb{Z}_5\} \quad i = \sqrt{-1}$$

- (i) How many elements are in $\mathbb{Z}_5[i]$
- (ii) The element $2+3i \in \mathbb{Z}_5[i]$ is a unit. Find its multiplicative inverse.
- (iii) The element $\alpha = 2+i \in \mathbb{Z}_5[i]$ is a zero-divisor. Find $\beta \in \mathbb{Z}_5[i]$ such that $\alpha \cdot \beta = 0$.
- (iv) Is $\mathbb{Z}_5[i]$ an integral domain, a field or neither.

Subring: Let R be a ring. A subset $S \subseteq R$ is a subring of R if S is a ring using the operations of R .

Ex: $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b \in \mathbb{Z}_{10} \right\}$ is a subring of $M(\mathbb{Z}_{10})$ ring.

$$A, B \in S \Rightarrow A+B \in S, A \cdot B \in S.$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in M(\mathbb{Z}_{10}) \text{ identity.}$$

$$A \in S \Rightarrow -A \in S.$$

$$\text{Multiplicative identity } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in M(\mathbb{Z}_{10})$$

Ex: for each $a \in (\mathbb{Z}_{10}, +, \cdot)$, find $-a$, and a^{-1} or explain why $-a$ or a^{-1} (or possibly both) does not exist.

- (i) $a = 3$, (ii) $a = 4$, (iii) $a = 1$, (iv) $a = 0$, (v) $a = 9$.

Ex. For each $\alpha \in M(\mathbb{Z}_6)$, find $-\alpha$ and α^{-1} or explain why $-\alpha$ or α^{-1} (or possibly both) does not exist.

(i) $\alpha = \begin{pmatrix} 4 & 7 \\ 1 & 7 \end{pmatrix}$ (ii) $\alpha = \begin{pmatrix} 4 & 6 \\ 1 & 7 \end{pmatrix}$ (iii) $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, (iv) $\alpha = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(v) $\alpha = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}$.

Quotient ring: $\underline{\mathbb{Z}_3[x] / \langle x^2 \rangle}$. (This is NOT a field).

Consider the poly. ring $\mathbb{Z}_3[x]$, and

$$\langle x^2 \rangle = \{ x^2 \cdot q(x) \mid q(x) \in \mathbb{Z}_3[x] \}. \quad (\text{all multiples of } x^2)$$

$$\text{Then } \mathbb{Z}_3[x] / \langle x^2 \rangle = \{ a(x) + \langle x^2 \rangle \mid a(x) \in \mathbb{Z}_3[x] \}.$$

Ex. $a(x) = x^9 + 2x^5 + x^3$
 $= x^2(x^7 + 2x^3 + x)$
 $\in \langle x^2 \rangle.$

$b(x) = 2x^7 + x^4 + 2x.$
 $\notin \langle x^2 \rangle.$ (Since $b(x)$ is not a multiple of x^2).

Ex. Let $\alpha(x), \beta(x) \in \mathbb{Z}_3[x]$ where.
 $\alpha(x) = x^9 + 2x^5 + x^3 + \underline{2x+1},$
 $\beta(x) = 2x+1.$

Then $\alpha(x) \neq \beta(x)$ in $\mathbb{Z}_3[x].$

But $\alpha(x) + \langle x^2 \rangle = \beta(x) + \langle x^2 \rangle,$

since $\alpha(x) - \beta(x) = x^9 + 2x^5 + x^3$
 $= x^2(x^7 + 2x^3 + x)$
 $\in \langle x^2 \rangle.$

Similar to.

$4378 + 5\mathbb{Z} = 3 + 5\mathbb{Z}$
in $\mathbb{Z}/5\mathbb{Z}.$

here $\underline{4378-3 \in 5\mathbb{Z}}$

Ex. Let $\alpha(x) = x^{13} + 2x^{10} + 2x^5 + \underline{x+2} \in \mathbb{Z}_3[x].$

Reduce the coset $\alpha(x) + \langle x^2 \rangle$. That is, we will find $\beta(x) \in \mathbb{Z}_3[x]$ of smallest degree such that;

$$\alpha(x) + \langle x^2 \rangle = \beta(x) + \langle x^2 \rangle.$$

Let $\beta(x) = x+2.$ Then

$$\alpha(x) - \beta(x) = x^{13} + 2x^{10} + 2x^5$$

$$= x^2(x^{11} + 2x^8 + 2x^3) \in \langle x^2 \rangle.$$

Therefore $\alpha(x) + \langle x^2 \rangle = \beta(x) + \langle x^2 \rangle$

Question: $\mathbb{Z}_3[x]/\langle x^2 \rangle$ is not a field. $= (x+2) + \langle x^2 \rangle$ in $\mathbb{Z}_3[x]/\langle x^2 \rangle$.

Note: (Imp): Every element of $\mathbb{Z}_3[x]/\langle x^2 \rangle$ can be reduced

to, $(ax+b) + \langle x^2 \rangle$ where $a, b \in \mathbb{Z}_3$.

Linear

Thus $\mathbb{Z}_3[x]/\langle x^2 \rangle = \{(ax+b) + \langle x^2 \rangle \mid a, b \in \mathbb{Z}_3\}$.

$|\mathbb{Z}_3| = 3$
 $(a, b) = ?$ } $3^2 = 9$ possibilities

additive inverse $\rightarrow$ $0 + \langle x^2 \rangle$
multiplicative inverse $\rightarrow$ $1 + \langle x^2 \rangle$

$x + \langle x^2 \rangle, (x+1) + \langle x^2 \rangle, (x+2) + \langle x^2 \rangle$

$2x + \langle x^2 \rangle, (2x+1) + \langle x^2 \rangle, (2x+2) + \langle x^2 \rangle\}$

(all these elements are distinct).

1) If possible,

$(2x+1) + \langle x^2 \rangle = (x+2) + \langle x^2 \rangle$

$\Rightarrow (2x+1) - (x+2) = x-1 \in \langle x^2 \rangle$, a contradiction.

2) Inverse of $(x+1) + \langle x^2 \rangle$ is $(2x+1) + \langle x^2 \rangle$.

$((x+1) + \langle x^2 \rangle) \cdot ((2x+1) + \langle x^2 \rangle)$
 $= (x+1)(2x+1) + \langle x^2 \rangle$
 $= (2x^2 + 3x + 1) + \langle x^2 \rangle$ ($3x = 0$ in $\mathbb{Z}_3[x]$)
 $= (2x^2 + 1) + \langle x^2 \rangle$
 $= 1 + \langle x^2 \rangle$. (since $(2x^2+1) - 1 \in \langle x^2 \rangle$).

3) $0 \neq x + \langle x^2 \rangle \in \mathbb{Z}_3[x]/\langle x^2 \rangle$

now $(x + \langle x^2 \rangle) \cdot (x + \langle x^2 \rangle) = x^2 + \langle x^2 \rangle$
 $= 0 + \langle x^2 \rangle$ (since $x^2 - 0 \in \langle x^2 \rangle$).

Therefore $x + \langle x^2 \rangle$ is not a unit, it is a zero divisor.

Therefore $\mathbb{Z}_3[x]/\langle x^2 \rangle$ is not a field.